

Lab 8 - Splunk SIEM Investigation

Windows Authentication Monitoring and Failed-Logon Detection

Attribute	Details
Splunk platform	Splunk Enterprise 10.4.3
Server	Ubuntu Server 24.04.5 LTS
Endpoint	Windows 11 VM + Universal Forwarder
Receiving port	9997/TCP
Primary source	Windows Security Event Log

1. Executive Summary

Splunk Enterprise was installed on a dedicated Ubuntu Server VM and configured to receive Windows telemetry through the Universal Forwarder. SPL searches identified successful and failed authentication events, a custom detection summarized bursts of Event ID 4625 failures, a scheduled alert was successfully triggered, and a dashboard summarized authentication activity.

2. Architecture and Configuration

Component	Role / Configuration
Ubuntu Server VM	Splunk Enterprise indexer/search head; Web 8000; receiving 9997.
Windows 11 VM	Universal Forwarder; Windows Security log enabled.
VMware NAT	Endpoint and indexer communicated on 192.168.180.0/24.
Windows event input	[WinEventLog://Security], index=main, renderXml=true.

3. Authentication Investigation

The Security log was searched for Event IDs 4624 and 4625. Because the XML sourcetype did not expose EventCode in the required context, SPL rex extraction was used to parse the EventID from raw XML.

Event	Meaning	Initial observed count
4624	Successful logon	16
4625	Failed logon	3

4. Failed-Logon Detection

Field	Result
Host	DESKTOP-ICS24PI
Source IP	127.0.0.1 - localhost
Logon Type	2 - interactive local logon
Failed attempts	3
First seen	2026-09-30 12:35:52.726
Last seen	2026-09-30 12:35:56.536

The test represented local interactive authentication failures and did not support a claim of remote brute-force activity.

5. Scheduled Alert and Dashboard

- Created the alert Multiple Failed Windows Logons.
- Scheduled execution every five minutes with trigger condition Number of Results > 0.
- Used Add to Triggered Alerts and verified successful firing during a controlled test.
- Built a Windows Authentication Monitoring dashboard with a successful-vs-failed logon chart and failed-logon summary table.

6. Skills Demonstrated

- Splunk Enterprise deployment
- Universal Forwarder configuration
- Windows Security log ingestion
- SPL and rex field extraction
- Detection engineering
- Scheduled alerting

- Dashboard construction
- Authentication analysis

7. Analyst Assessment

Classification: controlled authorized test. The lab demonstrated ingestion, search, detection, alerting, and visualization without mischaracterizing the localhost Logon Type 2 activity as a remote attack.